

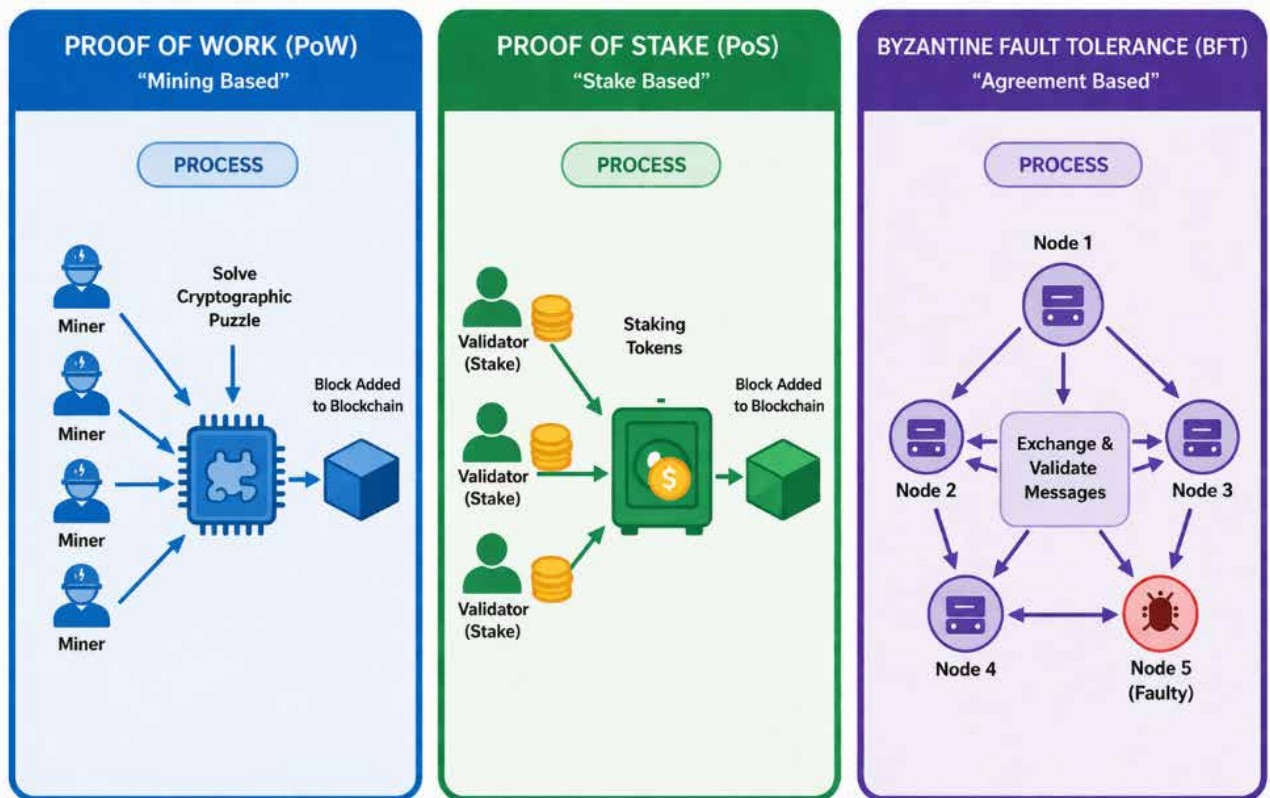


Figure 2: Blockchain consensus mechanisms: PoW vs PoS vs BFT

- Supports smart contracts and decentralised applications.
- Improved scalability and reduced energy consumption.

Hyperledger and Cosmos

- Use BFT-based consensus mechanisms.
- Designed for enterprise and permissioned environments.
- Provide fast transaction finality and high efficiency.
- Suitable for controlled networks with known participants.

Challenges and future scope

There are difficulties that all blockchain consensus protocols deal with including, but not limited to, lack of performance and scalability. While PoW and PoS can securely verify transactions, both are not as scalable as BFT-based approaches. However, BFT-based systems have their own set of challenges derived

from high communication overhead and the costs associated with maintaining several nodes.

Hybrid consensus models that combine, for example, Proof of Stake (PoS) and Byzantine Fault Tolerance

(BFT) are the future. Other types of sharding solutions that provide more expansive scalability, increased performance levels, and a more sustainable approach for blockchain networks are also on the cards. **END** 🐧

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